Scope 3 Emissions Estimation from Procurement Text

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Problem and Motivation

Scope 3 emissions, those generated across a company's supply chain, typically account for most of its total carbon footprint. The most common way to estimate them is the spend-based method: each purchase is assigned an industry code (NAICS) and multiplied by that industry's emission factor from the EPA's Supply Chain GHG Emission Factors. The bottleneck is classification. Analysts must manually map thousands of loosely worded purchase descriptions to industry codes, a process that is slow, costly, and inconsistent. With mandatory Scope 3 disclosure arriving under California's SB 253 and the EU's CSRD, many organizations will need to produce these estimates repeatedly and defensibly, often without dedicated sustainability staff.

Intended User and Task

The intended users are sustainability analysts and procurement teams preparing emissions reports. The task is: given a purchase record (free-text description and dollar amount), assign the correct NAICS-6 industry code, compute the associated emissions, and flag low-confidence items for human review.

Why Generative AI Is Appropriate

Purchase descriptions are short, inconsistent, and full of abbreviations, vendor jargon, and ambiguous terms, which makes keyword rules brittle. There are over 1,000 NAICS-6 codes, too many for a conventional classifier to learn well from limited labeled data. LLMs bring broad world knowledge of products and industries, can reason over code definitions provided through retrieval, and can explain their choices, which supports auditability. They also generalize to new descriptions without retraining.

Initial Definition of Success

Success will be measured on a held-out test set of about 500 federal contracts from USAspending.gov, which come labeled with NAICS codes. The primary metric is emissions-weighted error: the total difference in estimated kgCO2e between predicted and true codes, since misclassifications between industries with similar emission factors matter far less than those between dissimilar ones. Secondary metrics are classification accuracy at the 2-, 4-, and 6-digit NAICS levels and cost and latency per item. A successful system should clearly outperform a zero-shot baseline on emissions-weighted error, and its confidence scores should identify the items most likely to be wrong.

Preliminary System Direction

We will start with a zero-shot baseline in which an LLM directly predicts a NAICS-6 code from the description. We will then add retrieval, embedding NAICS definitions so the model chooses among the top candidate codes, followed by few-shot examples retrieved from labeled training data. Errors at each stage will be categorized (vague descriptions, goods versus services confusion, near-neighbor codes) to guide the next iteration. If time permits, we will extend the system with confidence-based abstention for human review, hierarchical classification, a comparison of small, large, and fine-tuned models, and robustness testing on terse corporate expense-style descriptions.